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United States Patent	12397926
Kind Code	B2
Date of Patent	August 26, 2025
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Vertiport system including transfer unit using rail moving device

Abstract

A vertiport system according to an embodiment may include a first area, a second area provided in a different position from the first area, and a transfer unit configured to transfer an aircraft between the first area and the second area, wherein the transfer unit may include a transfer rail connecting the first area and the second area, and a moving device configured to accommodate at least a portion of a landing gear of the aircraft and the moving device being configured to move along the transfer rail.

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Appl. No.: 18/611029

Filed: March 20, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20250171162 A1	May. 29, 2025

Foreign Application Priority Data

KR	10-2023-0167995	Nov. 28, 2023
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Publication Classification

Int. Cl.: B64F1/223 (20240101)

U.S. Cl.:

CPC **B64F1/223** (20240101);

Field of Classification Search

CPC: B61B (10/00); B61B (10/001); B61B (13/00); B64F (1/002); B64F (1/007); B64F (1/04); B64F (1/06); B64F (1/12); B64F (1/125); B64F (1/16); B64F (1/22); B64F (1/222); B64F (1/223); B64F (1/224); B64F (1/225); B64F (1/226); B64F (1/227); B64F (1/228); B64F (1/24); B64G (1/002); B64G (1/005); B64U (70/90); B64U (70/97); B64U (70/99); E01B (21/00); E01B (25/00); E01B (25/22); E01B (25/28); E04H (6/44)

USPC: 104/106; 104/107; 104/110; 104/139; 104/140; 244/116

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of Korean Patent Application No. 10-2023-0167995, filed on Nov. 28, 2023, which application is hereby incorporated herein by reference in its entirety.

TECHNICAL FIELD

(2) The present disclosure relates to a vertiport system including a transfer unit using a rail moving device.

BACKGROUND

(3) Recently, urban air mobility (UAM) for future transportation and traffic service systems is being developed. UAM may be an aircraft capable of vertical takeoff and landing flights, and along with the development of the UAM, the development of a vertiport, a takeoff and landing site of the UAM, is required.

(4) A vertiport has infrastructure or systems with support services and equipment for landing, ground handling, and take-off of manned or unmanned vertical takeoff and landing (VTOL) aircraft. The vertiport is generally considered to be installed on rooftops of high-rise buildings or large parks. The vertiport may have a smaller footprint with smaller landing and take-off regions, as compared to typical airports having typical runways.

(5) On the other hand, the UAM requires ground driving movement for various purposes, such as customers getting on or off a vehicle/charging/parking/hangar transfer within the vertiport, etc.

SUMMARY

(6) The present disclosure relates to a vertiport system including a transfer unit using a rail moving device.

(7) An embodiment of the present disclosure can provide a vertiport system capable of safely transferring an aircraft in a designed, selected, preset, or predetermined transfer trajectory using a rail moving device moving along a rail.

(8) According to an embodiment of the present disclosure, a vertiport system may include a first area, a second area provided in a different position from the first area, and a transfer unit provided to transfer an aircraft between the first area and the second area, wherein the transfer unit comprises a transfer rail connecting the first area and the second area and a moving device in which at least a portion of a landing gear of the aircraft is accommodated and which moves along the transfer rail.

(9) In an embodiment, the moving device may include a jig to which the landing gear is connected, a wheel rotatably connected to the jig, and a motor connected to the jig and providing driving force to rotate the wheel.

(10) In an embodiment, the jig may include a gear settling portion in which the landing gear is accommodated, and a wheel connection portion to which the wheel is connected, and the motor may be provided in the wheel connection portion and coupled to the wheel.

(11) In an embodiment, the vertiport system may further include a moving device controller configured to control an operation of the moving device, and the moving device may further include a communication controller configured to receive a signal from the moving device controller or transmit a signal to the moving device controller.

(12) In an embodiment, the moving device controller may generate a moving signal including at least one of whether the moving device is driven, a moving speed, and a moving direction and may transmit the moving signal to the communication controller.

(13) In an embodiment, the communication controller may be connected to the motor, and may

control driving of the motor based on the moving signal received from the moving device controller.

(14) In an embodiment, the wheel connection portion may be positioned below the gear settling portion, and the gear settling portion may include a settling groove having a tapered shape narrowing downwardly.

(15) In an embodiment, the gear settling portion may include a bottom surface on which the landing gear is accommodated, and a pair of guide surfaces extending obliquely outwardly from both sides of the bottom surface to an upward direction.

(16) In an embodiment, the guide surface may guide the landing gear to be aligned with the bottom surface in the process of settling the landing gear in the gear settling portion due to a movement of the aircraft.

(17) In an embodiment, the first area, the second area and the transfer rail may be provided on a floor of a port body, and a rail accommodating groove recessed to a designed, selected, or predetermined depth may be formed in the floor so that the transfer rail is positioned.

(18) In an embodiment, the rail accommodating groove may be recessed to a specified depth so that the bottom surface is positioned at the same height as the floor, in a state in which the moving device is accommodated on the transfer rail.

(19) In an embodiment, the rail accommodating groove may be recessed to a specified depth so that the bottom surface is positioned at a height lower than the floor by a specified length, in a state in which the moving device is accommodated on the transfer rail.

(20) In an embodiment, the landing gear of the aircraft may include a nose gear, a first main gear, and a second main gear, the moving device may include a first moving device on which the first main gear is accommodated, and a second moving device on which the second main gear is accommodated, and the transfer rail may include a first rail along which the first moving device moves, and a second rail along which the second moving device moves.

(21) In an embodiment, the first rail and the second rail have the same shape and are spaced apart from each other at a specified interval, and in a state in which the moving device is positioned on the transfer rail and the main gear is accommodated in the moving device, the nose gear may be in contact with a portion of the floor positioned between the first rail and the second rail.

(22) In an embodiment, the transfer rail may be formed to have a shape including a straight section and a curved section, and the moving device controller may control the first moving device and the second moving device so that the first moving device and the second moving device move at different speeds in the curved section.

(23) According to an embodiment of the present disclosure, a rail moving device may include a jig to which landing gear of an aircraft is connected, a wheel rotatably connected to the jig and moving along a rail, and a motor connected to the jig and providing driving force to rotate the wheel, where the jig may include a gear settling portion in which the landing gear is accommodated, and a wheel connection portion to which the wheel is connected, and the motor may be provided in the wheel connection portion and coupled to the wheel.

(24) In an embodiment, the wheel connection portion may extend downwardly from the gear settling portion, and the gear settling portion may include a settling groove having a tapered shape narrowing downwardly, and the landing gear may be accommodated in the settling groove.

(25) In an embodiment, the gear settling portion may include a bottom surface on which the landing gear is accommodated, and a pair of guide surfaces extending obliquely outwardly from both sides of the bottom surface to an upward direction.

(26) In an embodiment, the guide surface may guide the landing gear to be aligned with the bottom surface in the process of settling the landing gear in the gear settling portion.

(27) In an embodiment, the rail moving device may further include a communication controller connected to the motor and configured to control driving of the motor based on a moving signal received from an external controller, and the moving signal may include at least one of whether the

rail moving device is driven, a moving speed, and a moving direction.

(28) According to an embodiment of the present disclosure, by transferring an aircraft using a rail moving device moving along a rail, it may be possible to safely and easily transfer the aircraft in a designed, selected, or predetermined transfer trajectory within a narrow space.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The above and other features and advantages of the present disclosure can be more clearly understood from the following detailed description, taken in conjunction with the accompanying drawings, in which:

(2) FIG. 1 is a perspective view illustrating a vertiport system according to an embodiment of the present disclosure;

(3) FIG. 2 is a perspective view illustrating a state in which an aircraft is connected to a transfer unit of a vertiport system according to an embodiment of the present disclosure;

(4) FIG. 3 are perspective views illustrating a moving device of a vertiport system according to an embodiment of the present disclosure;

(5) FIG. 4 is a front view illustrating a state in which a main gear of an aircraft is accommodated in a moving device of a vertiport system according to an embodiment of the present disclosure;

(6) FIG. 5 are perspective views illustrating a state in which main gear members of various sizes are accommodated in a jig of a moving device in a vertiport system according to an embodiment of the present disclosure;

(7) FIGS. 6A, 6B, and 6C are perspective views illustrating an operation in which an aircraft is transferred by a transfer unit of a vertiport system according to an embodiment of the present disclosure; and

(8) FIG. 7 is a flowchart illustrating a method of controlling a vertiport system according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF ILLUSTRATIVE EMBODIMENTS

(9) Various changes and various embodiments of the present disclosure are possible, and specific example embodiments thereof will be described and illustrated in the drawings hereinafter.

However, the example embodiments are not intended for necessarily limiting the present disclosure. Ideas of the present disclosure can be extended to any alterations, equivalents, and substitutes besides the accompanying drawings.

(10) It can be understood that although the terms “first,” “second,” and the like, may be used herein to describe various elements, these elements are not necessarily limited by such terms. Such terms can be generally used merely to distinguish one element from another. For example, a first element can be termed a second element, and, similarly, a second element can be termed a first element, without departing from the scope of the present disclosure. The term “and/or” can encompass a combination of plural items or any one of the plural items.

(11) Terms used herein can be for the purpose of describing particular example embodiments only and are not intended to be necessarily limiting of the present disclosure. The singular can also include the plural unless specifically stated otherwise in the phrase. It can be further understood that the terms “comprises,” “comprising,” “includes” and/or “including,” when used herein, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

(12) Unless otherwise defined, terms including technical and scientific terms used herein can have a same meaning as commonly understood by one of ordinary skill in the art to which example embodiments of the present disclosure pertain. It can be further understood that terms, such as

those defined in commonly used dictionaries, can be interpreted as having meanings that are consistent with their meanings in the context of the relevant art.

(13) In the present specification, an aircraft may refer to a mobility vehicle that may move by flying in the sky. In addition to referring to helicopters, drones, fixed-wing airplanes, and the like, an aircraft may include vehicles that can move on the ground using wheels and then fly with the wheels separated from the ground. In addition, an aircraft may include a manned aircraft and an unmanned aircraft. The manned aircraft may include an airplane that can be operated by autonomous flight, in addition to an airplane controlled by a pilot.

(14) Hereinafter, embodiments of the present disclosure will be described with reference to the attached drawings.

(15) FIG. 1 illustrates a vertiport system **100** according to an embodiment of the present disclosure. FIG. 2 illustrates a state in which an aircraft **200** is connected to a transfer unit **120** of the vertiport system **100** according to an embodiment of the present disclosure.

(16) FIG. 1 is a view schematically illustrating a vertiport system **100** according to an embodiment. FIG. 2 is an enlarged view of part A illustrated in FIG. 1.

(17) The vertiport system **100** according to an embodiment is a takeoff and landing site (e.g., a vertical takeoff and landing airfield or station) for the aircraft **200** capable of vertical takeoff and landing (VTOL), and may include infrastructure configured to receive an aircraft from a flight, reset the aircraft for a subsequent flight, and allow the aircraft to depart for a subsequent flight. The aircraft **200** may be air mobility including Urban Air Mobility (UAM) and Advanced Air Mobility (AAM), and the type of aircraft **200** is not particularly limited.

(18) The vertiport system **100** refers to various facilities providing space for vertical takeoff and landing of aircraft, and may be understood as including Verti-hub, Verti-port, or Verti-stop, classified according to size. For example, a Verti-hub can be the largest UAM takeoff and landing site, and may enable large-scale transfer of surrounding traffic of airports or the like, may have support an infrastructure such as charging and maintenance, and may allow multiple UAM vehicles to park after their operation ends. A Verti-port is a smaller takeoff and landing site than the Verti-hub, and may have a vehicle support infrastructure (charging, maintenance, etc.) and may also provide passenger convenience facilities. A Verti-stop is smaller than the Verti-port and may be a small takeoff and landing site with one or two aprons.

(19) Referring to FIGS. 1 and 2, the vertiport system **100** according to an embodiment may include a port body **110**, a transfer unit **120**, a control unit **150**, and a controller **160**.

(20) The port body **110** may be equipped to allow the aircraft **200** to take off or land, stop, or move. For example, the port body **110** may be equipped with components and/or auxiliary facilities for takeoff, landing, stopping, and movement of the aircraft **200**.

(21) The port body **110** may include a takeoff and landing area **111** in which the aircraft **200** takes off or lands, and a transfer area **113** for stopping the aircraft **200** or moving the aircraft **200** to another area provided in the port body **110** (or the vertiport system **100**). One or more of each of the takeoff and landing area **111** and the transfer area **113** may be provided. On the other hand, the takeoff and landing area **111** and the transfer area **113** are examples of separate areas provided for the aircraft **200** to be transferred, located, or stopped, and may also be referred to as a first area and a second area, according to various embodiments.

(22) The takeoff and landing area **111** and the transfer area **113** may be provided at different positions in the port body **110**. The takeoff and landing area **111** may be provided in a partial region of the port body **110**, and the transfer area **113** may be provided in another area of the port body **110** separated from the takeoff and landing area **111**. For example, the port body **110** may be configured in a layout in which the takeoff and landing area **111** is positioned on one side of a base area **115**, and the transfer area **113** is positioned on the other side of the base area **115**.

(23) The aircraft **200** may take off or land vertically in the takeoff and landing area **111**. The aircraft **200** that landed in the takeoff and landing area **111** may be transferred to the transfer area

113 through the transfer unit **120**. Additionally, the aircraft **200** located in the transfer area **113** may be transferred to the takeoff and landing area **111** through the transfer unit **120**.

(24) The takeoff and landing area **111** and the transfer area **113** may be provided in the form of a circular platform installed in the port body **110** (or the base area **115**). According to various embodiments, at least one of the takeoff and landing area **111** and the transfer area **113** may be rotatable with respect to the port body **110**. The circular platform of the takeoff and landing area **111** and/or the transfer area **113** may be provided to physically rotate. For example, at least one of the takeoff and landing area **111** and the transfer area **113** may be provided with a motor for rotating the circular platform, and the circular platform may be coupled to be rotated by a motor. The takeoff and landing area **111** and the transfer area **113** may be turntables.

(25) The transfer area **113** may be rotatable with respect to the port body **110**. For example, the transfer area **113** (i.e., the circular platform of the transfer area **113**) may rotate with respect to the base area **115** and the takeoff and landing area **111** of the port body **110**. An aircraft **200** stopped in the transfer area **113** may rotate with the transfer area **113** as the transfer area **113** rotates, and may adjust a direction in which the aircraft **200** located in the transfer area **113** is headed based on the rotation of the transfer area **113**.

(26) The transfer area **113** may be an area in which the aircraft **200** transferred from the takeoff and landing area **111** temporarily stops before being transferred to another area, or may be an area in which the aircraft **200** transferred from other areas temporarily stops before being transferred to the takeoff and landing area **111**. Additionally, in another example, the transfer area **113** may be an area in which the aircraft **200** temporarily stops for boarding or disembarking boarded passengers and/or loaded cargo, but the present disclosure is not necessarily limited thereto.

(27) The transfer area **113** may be configured to transfer the aircraft **200** to a maintenance area, a charging area, a boarding and disembarking area, a transfer area, and a containment area provided in the vertiport system **100** (or a smart building including the vertiport system **100**). For example, FIG. **1** may be understood as a view schematically illustrating a rooftop of a building including the vertiport system **100**, and the transfer area **113** (i.e., the circular platform of the transfer area **113**) may be provided to enable ascending and descending in a building, such as an elevator.

(28) The transfer unit **120** may transfer/move the aircraft **200**. For example, the transfer unit **120** may be provided to move the aircraft **200** between the takeoff and landing area **111** and the transfer area **113**. The takeoff and landing area **111** and the transfer area **113** are examples, and the transfer unit **120** is not necessarily limited to transferring the aircraft **200** between the takeoff and landing area **111** and the transfer area **113**, and according to various embodiments, the transfer unit **120** may be provided to transfer the aircraft **200** between the first area and the second area, which are separated and recognized from each other.

(29) The transfer unit **120** may include a transfer rail **130** and a moving device **140**. The transfer rail **130** may be provided in the port body **110** and may connect the takeoff and landing area **111** and the transfer area **113**. The moving device **140** may move between the takeoff and landing area **111** and the transfer area **113** along the transfer rail **130**.

(30) Referring to FIG. **2**, the transfer unit **120** may be provided to tow and transfer the aircraft **200** by settling at least a portion of the landing gear of the aircraft **200** in the moving device **140** and allowing the moving device **140** to move along the transfer rail **130**.

(31) The aircraft **200** can be air mobility capable of vertical takeoff and landing, and may include an airframe **210** (or fuselage) capable of boarding passengers and/or loading cargo, a wing portion **220** coupled to an upper portion of the airframe **210** and provided with a rotor for generating lifting force, and a plurality of landing gear members **230**, **240** coupled to the lower part of the aircraft **200** and configured to support takeoff, landing, and ground running of the aircraft **200**. The plurality of landing gear members may include a nose gear **230** positioned in the front and two main gear members **240** located in the rear. For example, the main gear **240** may include a first main gear **241** positioned on the right side with respect to the airframe **210** and a second main gear

242 positioned on the left side with respect to the airframe **210**.

(32) The transfer rail **130** may be installed in the port body **110**. For example, the transfer rail **130** may be installed on a floor **117** of the port body **110** so that the transfer rail **130** is positioned between the takeoff and landing area **111** and the transfer area **113**. The transfer rail **130** may extend from the takeoff and landing area **111** to the transfer area **113**. The transfer rail **130** may be connected to the takeoff and landing area **111** and the transfer area **113** so that the aircraft **200** positioned in the takeoff and landing area **111** or the transfer area **113** may move and the main gear **240** may be accommodated in the moving device **140** arranged on the transfer rail **130**.

(33) The transfer rail **130** may be formed to have a shape including at least one of a straight section and a curved section. According to the embodiment illustrated in FIG. 1, the transfer rail **130** may extend from the takeoff and landing area **111** in a straight section and then may extend to the transfer area **113** while being deformed into the curved section. However, the shape and/or structure of the transfer rail **130** is not limited to the illustrated embodiment, and may be modified in various manner in response to the locations of each area in which the aircraft **200** is to be transferred. For example, the transfer rail **130** may be composed of only straight sections.

(34) The transfer rails **130** may be provided as a pair so that two moving devices **140** may run/move while supporting two main gear members **240** of the aircraft **200**, respectively. The transfer rail **130** may include a first rail **131** on which a first moving device **140a** moves and a second rail **133** on which a second moving device **140b** moves.

(35) Referring to FIG. 2, a rail accommodating groove **119** in which the transfer rail **130** is installed may be formed in the port body **110**, and the transfer rail **130** may be positioned in the rail accommodating groove **119**. The rail accommodating groove **119** may be formed by recessing at least a portion of the floor **117** into a shape corresponding to the transfer rail **130**. The rail accommodating groove **119** may include a first accommodating groove **119-1** in which the first rail **131** is installed, and a second accommodating groove **119-2** in which the second rail **133** is installed.

(36) The moving device **140** may be connected to and/or fastened to the transfer rail **130** to move along the transfer rail **130**. The moving device **140** may transfer the aircraft **200** between the takeoff and landing area **111** and the transfer area **113** by moving along the transfer rail **130** in which the main gear **240** of the aircraft **200** is accommodated on at least a portion of the moving device **140**. For example, the moving device **140** may be referred to as a railcar.

(37) The moving device **140** may move along the transfer rail **130** based on a control signal transmitted from a moving device controller **161**. For example, the moving device **140** may receive moving information including whether the moving device **140** is driven, a moving speed, a moving direction, or any combination thereof, from the moving device controller **161** through communication, etc., and may operate based on the moving information.

(38) The moving device **140** may be comprised of two moving devices **140** so that the two main gear members **240** of the aircraft **200** are respectively accommodated. The moving device **140** may include a first moving device **140a** in which the first main gear **241** is accommodated and moves along the first rail **131**, and a second moving device **140b** in which the second main gear **242** is accommodated and moves along the second rail. The first moving device **140a** and the second moving device **140b** may have substantially the same structure, shape, and configuration.

(39) In a state where the main gear **240** of the aircraft **200** is accommodated in the moving device **140**, the nose gear **230** of the aircraft **200** may be in contact with a portion of the floor **117** positioned between the first accommodating groove **119-1** (or first rail **131**) and the second accommodating groove **119-2** (or second rail **133**). The nose gear **230** may move while rotating on the floor **117** when the moving device **140** moves along the transfer rail **130**, thus transferring the aircraft **200**.

(40) The first moving device **140a** and the second moving device **140b** may be driven by receiving separate control signals from the moving device controller **161**. The first moving device **140a** and

the second moving device **140b** may move at the same speed or may move at different speeds, depending on a section of the transfer rail **130**. For example, in a straight section of the transfer rail **130**, the first moving device **140a** and the second moving device **140b** may move in the same direction at the same speed, and in a curved section of the transfer rail **130**, the first moving device **140a** and the second moving device **140b** may move in the same direction at different speeds.

(41) For an embodiment, the specific configuration of the moving device **140** and the structure in which the main gear **240** is accommodated in the moving device **140** will be described in more detail with reference to FIGS. **3** and **4** below.

(42) The control unit **150** serves to manage and monitor the vertiport system **100** in an integrated manner and may control takeoff, landing, stopping, and transfer of the aircraft **200**. The control unit **150** may be positioned adjacent to the takeoff and landing area **111**, and may provide control information on take-off, landing, and transfer of the aircraft **200** to the controller **160** and/or ground personnel in comprehensive consideration of an operating environment in a control area.

(43) The controller **160** may control an activation and an operation of at least a portion of the vertiport system **100**. The controller **160** may include a moving device controller **161** controlling the moving device **140** of the transfer unit **120**, and a platform controller **162** controlling the rotation of the transfer area **113** and/or the takeoff and landing area **111**.

(44) The moving device controller **161** may control the driving and movement of the moving device **140**. For example, the moving device controller **161** may control at least one of whether each of the first moving device **140a** and the second moving device **140b** is driven, the moving direction, and the moving speed, or any combination thereof, for example. The moving device controller **161** may control the moving speeds of the first moving device **140a** and the second moving device **140b** to be different from each other so that the moving device **140** may move the curved section of the transfer rail **130**. The moving device controller **161** may control the moving device **140** by transmitting and receiving control signals through a communication controller (e.g., the communication controller **148** in FIG. **3**) provided in the moving device **140**.

(45) The platform controller **162** may control a rotational operation of a platform in the transfer area **113**. However, the platform controller **162** is not limited to rotation control of the transfer area **113**, and according to various embodiments, when the takeoff and landing area **111** is rotatably provided, the platform controller **162** may be provided to control the rotational operation of the platform of the takeoff and landing area **111**.

(46) FIG. **1** schematically illustrates a vertiport system **100** according to an embodiment, and the configurations of the vertiport system **100** are not limited to the illustrated embodiment, and the vertiport system **100** may further include other components (such as a takeoff and landing guide unit, a walking area, etc.) according to various embodiments.

(47) Furthermore, according to the embodiment illustrated in FIG. **1**, the vertiport system **100** may include one takeoff and landing area **111** and one transfer area **113**, but the present disclosure is not limited thereto. According to various embodiments, the vertiport system **100** may include two or more of at least one of the takeoff and landing area **111** and the transfer area **113**. For example, the vertiport system **100** may include one takeoff and landing area **111** and a plurality of transfer areas **113**, and the transfer rail **130** may be provided to connect the plurality of transfer areas **113** and the takeoff and landing area **111**. In this case, the transfer rail **130** may extend from the takeoff and landing area **111**, and at least a portion thereof may be branched to be connected to the plurality of transfer areas **113**, and a rail changer configured to change or convert a moving direction of the moving device **140** may be provided at a branch point corresponding to a position in which the aircraft **200** is to be transferred.

(48) FIG. **3** illustrates a moving device **140** of the vertiport system **100** according to an embodiment of the present disclosure. FIG. **4** illustrates a state in which a main gear **240** of an aircraft **200** is accommodated in the moving device **140** of the vertiport system **100** according to an embodiment of the present disclosure.

(49) FIG. 3 is a view illustrating a perspective view of the moving device **140** as seen from the left and a perspective view of the moving device **140** as seen from the right, respectively. FIG. 4 is an enlarged view of part B illustrated in FIG. 2.

(50) FIG. 4 illustrates a first moving device **140a** in which a first main gear **241** of the aircraft **200** is accommodated and a first rail **131** in a first accommodating groove **119-1** in which the first moving device **140a** is accommodated, but the structure illustrated in FIG. 4 may be equally applied to a second main gear **242**, a second moving device **140b**, a second accommodating groove **119-2**, and a second rail **133**, for example. Hereinafter, in describing FIG. 4, the components are not divided into 'first' and 'second', and will be described as a rail accommodating groove **119**, a transfer rail **130**, a moving device **140**, and a main gear **240**, collectively.

(51) Referring to FIGS. 3 and 4, the moving device **140** according to an embodiment may be equipped with independent driving equipment and communication equipment and may move along the transfer rail **130**, and be configured so that the main gear **240** of the aircraft **200** is accommodated therein.

(52) The moving device **140** may include a jig **141**, a wheel **144**, a motor **145**, and a communication controller **148**. The moving device **140** may be provided in a form in which the wheel **144**, the motor **145**, and the communication controller **148** are positioned and mounted in the jig **141**. Although the embodiment of FIG. 3 shows one wheel **144** per moving device **140**, in other embodiments, any number of wheels (driven and undriven) may be incorporated per moving device **140**.

(53) The jig **141** may be configured so that the main gear **240** of the aircraft **200** to be transferred is accommodated therein, and may support the wheels **144** and other components (e.g., a motor **145** and a communication controller **148**) of the moving device **140**.

(54) The jig **141** may include a gear settling portion **142** in which the main gear **240** is accommodated, and a wheel connection portion **143** that extends from the gear settling portion **142** and to which the wheel **144** is connected. The gear settling portion **142** may form an upper portion of the jig **141**, and the wheel connection portion **143** may form a lower portion of the jig **141**.

(55) The gear settling portion **142** may be provided with a settling groove **1423** that is concavely formed so that the main gear **240** is accommodated. The settling groove **1423** may guide the settling of the main gear **240**, and may be formed in a tapered shape to enable the settling of the main gear **240** of various sizes.

(56) The gear settling portion **142** may include a bottom surface **1421**, parallel to the floor **117** and the transfer rail **130**, and a pair of guide surfaces **1422** that extend obliquely outwardly from two sides of the bottom surface **1421** in an upward direction. For example, the settling groove **1423** may be formed by a concave space formed by the bottom surface **1421** and the pair of guide surfaces **1422**.

(57) The bottom surface **1421** may be formed to have a width in a direction parallel to a central axis of a wheel of the main gear **240**, and the guide surface **1422** may extend from both edges of the bottom surface **1421** facing in a direction parallel to the central axis of the wheel. A width of the bottom surface **1421** may be greater than or substantially equal to a width of the wheel **144**. The width of the wheel **144** can denote a length in a direction of the central axis.

(58) The guide surfaces **1422** may guide the main gear **240** to be aligned in a given position. For example, when the aircraft **200** moves to the moving device **140**, in a case in which the wheel of the main gear **240** does not move by being aligned with the bottom surface **1421** and moves while contacting one or both of the guide surface **1422**, the guide surface **1422** is formed to be inclined in the direction of gathering toward the bottom surface **1421**, so that the wheel is aligned to be stably accommodated on the bottom surface **1421**.

(59) The main gear **240** mounted in the gear settling portion **142** may be fixed using a tiedown strap and/or using a brake device (e.g., a parking brake) of the main gear **240** itself. For example, in a state in which the main gear **240** is accommodated in contact with the bottom surface **1421** of the

gear settling portion **142**, the main gear **240** may be fixed so as not to rotate or move on the bottom surface **1421** by engaging a parking brake.

(60) An upward direction is a direction in which the bottom surface **1421** faces and indicates a direction oriented from the wheel connection portion **143** to the gear settling portion **142**, and a downward direction, opposite to the upward direction, is a direction oriented from the gear settling portion **142** to the wheel connection portion **143**.

(61) The wheel connection portion **143** may extend from a lower portion of the gear settling portion **142** to connect the wheel **144**. The wheel connection portion **143** may support the motor **145** and the communication controller **148** along with the wheel **144**. For example, the wheel **144**, motor **145**, and communication controller **148** may be mounted in the wheel connection portion **143** to form one assembly with the jig **141**.

(62) The wheel connection portion **143** may include a first connection portion **1431** and a second connection portion **1432** extending to face each other. The first connection portion **1431** and the second connection portion **1432** may be formed to be spaced apart so that the wheel **144** is positioned therebetween. For example, the first connection portion **1431** and the second connection portion **1432** may face both ends of the wheel **144**, respectively.

(63) The motor **145** may be positioned in the wheel connection portion **143**. For example, a motor body **146** of the motor **145** may be connected to one of the first connection portion **1431** and the second connection portion **1432** of the wheel connection portion **143**, and a shaft **147** of the motor **145** extending from the motor body **146** may penetrate through the other one of the first connection portion **1431** and the second connection portion **1432**.

(64) The communication controller **148** may be positioned in the wheel connection portion **143**. For example, the communication controller **148** may be positioned on one of the first connection portion **1431** and the second connection portion **1432** in which the motor body **146** is positioned.

(65) The wheel **144** may be rotatably connected to the jig **141**. For example, the wheel **144** may be rotatably connected to the wheel connection portion **143** of the jig **141**. The wheel **144** may be positioned between the first connection portion **1431** and the second connection portion **1432** and may be rotatably connected to the wheel connection portion **143** through the motor **145**. For example, the wheel **144** may be coupled to the motor **145** penetrating through the first connection portion **1431** and the second connection portion **1432**, and may be connected to the wheel connection portion **143** via the motor **145**. The wheel **144** may be rotated by driving force of the motor **145**. The wheel **144** may be connected to the shaft **147** of the motor **145**, and may be rotated together with the shaft **147** when the shaft **147** rotates.

(66) The wheel **144** may be accommodated on the transfer rail **130** and may move while rotating along the transfer rail **130**. For example, the wheel **144** may perform rotational motion and translational motion with respect to the transfer rail **130**. The wheel **144** may include a groove **1441** in which at least a portion of the transfer rail **130** is positioned so that the wheel **144** is stably mounted on the transfer rail **130** and rotates. The groove **1441** may be formed with a width corresponding to the transfer rail **130**. For example, when the moving device **140** is mounted on the transfer rail **130**, the transfer rail **130** can be partially accommodated in the groove **1441** of the wheel **144**, thereby preventing separation of the wheel **144** from the transfer rail **130**.

(67) According to the illustrated embodiment, a groove **1441** is formed in the wheel **144** and the transfer rail **130** has a protruding shape corresponding to the groove **1441**, but this is an example, and these shapes may be deformed opposite to each other. For example, the transfer rail **130** and the wheel **144** may be deformed so that the transfer rail **130** has a concave groove shape and the wheel **144** is partially accommodated in the transfer rail **130** having the concave shape.

(68) The motor **145** may provide driving force for moving the moving device **140**. The motor **145** may be positioned on the wheel connection portion **143**. The motor **145** may rotate the wheel **144** by transmitting the driving force to the wheel **144**. The motor **145** may be coupled to the wheel connection portion **143** and the wheel **144**. For example, the motor **145** may be provided so that a

portion thereof is coupled to the wheel connection portion **143** and the other portion thereof is coupled to the wheel **144**.

(69) The motor **145** may include a motor body **146** for generating driving force and a shaft **147** which can be connected to the motor body **146** and to which the wheel **144** can be connected. The motor body **146** and the shaft **147** may be coupled to the wheel connection portion **143**. The motor body **146** may be fixedly coupled to the wheel connection portion **143**, and the shaft **147** may be rotatably coupled to the wheel connection portion **143**. At least a portion of the shaft **147** may be inserted into the wheel **144**, and the wheel **144** may be provided to rotate through the rotation of the shaft **147**.

(70) According to the embodiment illustrated in FIG. 3, the motor body **146** may be coupled to the first connection portion **1431**, and the shaft **147** may extend from the motor body **146** and penetrate through the second connection portion **1432**. One end (e.g., a left end based on FIG. 3) of the shaft **147** may be coupled to the motor body **146** to receive the driving force from the motor body **146**, and the other end (e.g., a right end based on FIG. 3) of the shaft **147** may be rotatably coupled to the second connection portion **1432**. For example, a bearing **149** for rotating the shaft **147** may be positioned between an outer circumferential surface of the other end of the shaft **147** penetrating through the second connection portion **1432**, and the second connection portion **1432**.

(71) The communication controller **148** may transmit/receive a signal for control and/or operation of the moving device **140**. For example, the moving device **140** may receive control information or a control signal from the moving device controller **161** through the communication controller **148**, and may generate driving force of the motor **145** based on the transmitted information or signal.

(72) The communication controller **148** may be positioned in the wheel connection portion **143**. The communication controller **148** may be coupled to the motor **145**. For example, the communication controller **148** may be coupled to the motor body **146**. The communication controller **148** may be provided to be detachably attached to the wheel connection portion **143**. For example, the communication controller **148** may be detachably coupled to the motor **145**. A coupling position of the communication controller **148** is not limited to the motor body **146**.

(73) The communication controller **148** may be electrically connected to the motor **145**. For example, the communication controller **148** may be electrically connected to the motor body **146** to transmit the control signal provided from the moving device controller **161** to the motor **145**.

(74) The communication controller **148** may be provided to be detachably attached to the wheel connection portion **143**. For example, the communication controller **148** may be detachably coupled to the motor **145**.

(75) Referring to FIG. 4, a rail accommodating groove **119** may be formed by being recessed from the port body **110** to a designed, selected, or predetermined depth, and the transfer rail **130** may be positioned on a bottom surface of the rail accommodating groove **119**. The moving device **140** may be accommodated on the transfer rail **130** while being accommodated within the rail accommodating groove **119**.

(76) The rail accommodating groove **119** may be recessed from the floor **117** by a depth sufficient to be accommodated in the gear settling portion **142** of the moving device **140** by moving the aircraft **200** positioned in the takeoff and landing area **111** or the transfer area **113**. For example, the takeoff and landing area **111** and the transfer area **113** may be positioned at the same height as the floor **117**, and the rail accommodating groove **119** may be recessed to a depth at which the bottom surface **1421** of the gear settling portion **142** may be positioned at the same height as the floor **117** or may be positioned at a height lower than the floor **117** by a specified length L in a state in which the moving device **140** is accommodated on the transfer rail **130**.

(77) According to the embodiment illustrated in FIG. 4, the rail accommodating groove **119** may be formed at a depth such that the bottom surface **1421** of the gear settling portion **142** is positioned below the floor **117** by a specified length L in a state in which the moving device **140** is accommodated on the transfer rail **130**. As the specified length L is reduced, the main gear **240** of

the aircraft **200** positioned on the floor **117** moves to stably and smoothly to perform an operation of being accommodated on the gear settling portion **142**.

(78) As described above, according to various embodiments, the specified length **L** may be substantially zero so that the bottom surface **1421** of the gear settling portion **142** of the moving device **140** accommodated on the transfer rail **130** is positioned on the same line as the floor **117**.

(79) On the other hand, when the moving device **140** is accommodated on the transfer rail **130**, the bottom surface **1421** of the gear settling portion **142** does not necessarily have to be positioned at the same height as the floor **117** or at a lower height than the floor **117**, and according to various embodiments, when the moving device **140** is accommodated on the transfer rail **130**, the bottom surface **1421** of the gear settling portion **142** may be positioned to be higher than the floor **117** by a predetermined length.

(80) FIG. 5 illustrates a state in which the main gear **240** of various sizes is accommodated in the jig **141** of the moving device **140** in the vertiport system **100** according to an embodiment of the present disclosure.

(81) The moving device **140** illustrated in FIG. 5 is the same as the moving device **140** described with reference to FIGS. 3 and 4, and duplicate descriptions thereof will be omitted below.

(82) Referring to FIG. 5, the moving device **140** according to an embodiment may include a jig **141**, a wheel **144**, a motor **145**, and a communication controller **148**, and the jig **141** may include a gear settling portion **142** and a wheel connection portion **143**.

(83) The jig **141** of the moving device **140** may be provided with a taper-shaped settling groove **1423** so that the gear settling portion **142** may easily accommodate and align the main gear **240** (especially wheels) of various sizes.

(84) Comparing left and right images of FIG. 5, even when a wheel size of the main gear **240** is relatively small or large, an alignment of the main gear **240** may be guided by the guide surface **1422**, so that the wheel of the main gear **240** may be easily accommodated on the gear settling portion **142**.

(85) For example, in a state in which the wheel size of the main gear **240** is relatively large, when the aircraft **200** is moved to connect/install the aircraft **200** to the moving device **140**, it may be somewhat difficult to move the aircraft **200** with the wheels of the main gear **240** accurately aligned on the bottom surface **1421**. However, as inclined guide surfaces **1422** are provided on both sides of the bottom surface **1421**, the main gear **240** may be aligned by the guide surface **1422** even if the wheels of the main gear **240** are not moved to a given position.

(86) FIGS. 6A, 6B and 6C illustrate an operation in which the aircraft **200** is transferred by the transfer unit **120** of the vertiport system **100** according to an embodiment of the present disclosure.

(87) FIGS. 6A, 6B and 6C illustrate an operation in which the aircraft **200** lands in the takeoff and landing area **111** and then is transferred to the transfer area **113** by the transfer unit **120**.

Hereinafter, in describing FIGS. 6A, 6B and 6C, FIGS. 1 to 4 are also referred to together.

(88) Referring to FIG. 6A, the aircraft **200** lands in the takeoff and landing area **111** under the control of the control unit **150**. After the aircraft **200** lands in the takeoff and landing area **111**, the aircraft **200** moves to a position to be connected to the moving device **140**. The aircraft **200** can be moved to a position to be connected to the moving device **140** by ground personnel, and the main gear **240** can be accommodated in the gear settling portion **142** of the moving device **140**.

(89) For example, ground personnel can manually transfer the aircraft **200** and accommodate the two main gear members **240** in the gear settling portions **142** of each of the first moving device **140a** and the second moving device **140b**. A nose gear **230** of the aircraft **200** can rotate and move while coming into contact with the floor **117** between the first rail **131** and the second rail **133**.

According to various embodiments, the aircraft **200** may be manually transferred by ground personnel pulling a tow-bar connected to the nose gear **230** of the aircraft **200**. However, a manual transfer method of the aircraft **200** is not particularly limiting for an embodiment.

(90) Referring to FIG. 6B, the main gear **240** may be accommodated in the gear settling portion

142 and a parking brake of the main gear **240** may be engaged to secure the main gear **240** to the moving device **140**, and then, the aircraft **200** may be transferred by driving the moving device **140**. The moving device **140** can move the aircraft **200** to the transfer area **113** by moving along the transfer rail **130**. The moving device **140** can move along the transfer rail **130** by receiving control information from the moving device controller **161** and generating driving force of the motor **145**. (91) The transfer rail **130** provided between the takeoff and landing area **111** and the transfer area **113** may include a straight section and a curved section. When the moving device **140** moves in the straight section, the first moving device **140a** and the second moving device **140b** can move in the same direction at the same speed. When the moving device **140** moves in the curved section, the first moving device **140a** and the second moving device **140b** can move in the same direction at different speeds. For example, based on the shape of the transfer rail **130** illustrated in FIG. 6B, the second moving device **140b** moving along the second rail **133** in the curved section may move at a relatively slower speed than the first moving device **140a** moving along the first rail **131**, and may stably transfer the aircraft **200** even in the curved section.

(92) When the moving device **140** moves in the straight section, the moving device controller **161** may generate and transmit a control signal for moving the first moving device **140a** and the second moving device **140b** in the same direction at the same speed, and when the moving device **140** moves in the curved section, the moving device controller **161** may generate and transmit the control signal for moving the first moving device **140a** and the second moving device **140b** in the same direction at different speeds. Whether to control the speed of the first moving device **140a** or the second moving device **140b** relatively slowly may be determined based on a direction of the curved section. For example, unlike the drawing, when a section is bent from the straight section to the left (e.g., a side in which the first main gear **241**, the first rail **131**, or the first moving device **140a** is located), a speed of the first moving device **140a** moving on the first rail **131** is controlled to be slower than a speed of the second moving device **140b** moving on the second rail **133**.

(93) Additionally, whether to control the speeds of the first moving device **140a** and the second moving device **140b** to a different degree may be determined based on a curvature of the curved section. For example, when the curvature of the curve section is relatively large, a speed difference between the first moving device **140a** and the second moving device **140b** increases, and when the curvature of the curved section is relatively small, the speed difference between the first moving device **140a** and the second moving device **140b** decreases.

(94) According to various embodiments, when the moving device **140** moves along the transfer rail **130**, steering of the nose gear **230** may be controlled by a pilot of the aircraft **200**. For example, the main gear **240** may move the transfer rail **130** in the curved section by the moving device **140** that moves along the transfer rail **130**. However, because the nose gear **230** does not move along a designated path, steering in response to the curved section may be required, and the steering of the nose gear **230** may be performed by a pilot manipulating a steering device provided in a cockpit of the aircraft **200**.

(95) Referring to FIG. 6C, after the aircraft **200** is transferred to a position adjacent to the transfer area **113**, the aircraft **200** can be moved to the transfer area **113** by ground personnel. For example, when the moving device **140** has moved to an end of the transfer rail **130** connected to the transfer area **113**, after releasing the parking brake of the main gear **240**, the ground personnel may manually transfer the aircraft **200** and locate the aircraft **200** in the transfer area **113**. After the aircraft **200** is located in the transfer area **113**, a direction in which the aircraft **200** faces may be adjusted by rotating the transfer area **113** as necessary.

(96) FIG. 7 illustrates a control method of a vertiport system **100** according to an embodiment of the present disclosure.

(97) FIG. 7 is a flow chart illustrating a control method of the vertiport system **100** previously described with reference to FIGS. 1 to 4, and some operations of FIG. 7 may be performed by at least some components of the vertiport system **100** illustrated in FIGS. 1 to 4.

(98) Hereinafter, in describing FIG. 7, FIGS. 1 to 4 will be referred to together, and the content overlapping the previous description can be omitted.

(99) Referring to FIG. 7, a control method of the vertiport system **100** according to an embodiment may include an operation in which the aircraft lands (operation **S310**), an operation of instructing a control event (operation **S320**), an operation of connecting the main gear to the moving device (operation **S330**), an operation of driving the moving device (operation **S340**), an operation of confirming a position of the aircraft (operation **S350**), an operation of moving the aircraft to the floor (operation **S360**), and an operation of driving a turntable (operation **S370**).

(100) In the operation in which the aircraft lands (operation **S310**), the aircraft **200** may land in the takeoff and landing area **111**.

(101) In the operation of instructing a control event (operation **S320**), the control unit **150** may comprehensively consider an environment within a control area and deliver control information and instructions for the transfer of the aircraft **200** to the controller **160** and ground personnel.

(102) In the operation of connecting the main gear (operation **S330**), the ground personnel may manually move the aircraft **200** to a position of the moving device **140** mounted on the transfer rail **130** and accommodate the main gear **240** of the aircraft **200** in the moving device **140**. For example, the aircraft **200** can move by rotation of the main gear **240** while being positioned on the floor **117** of the takeoff and landing area **111**, so that the main gear **240** can be accommodated in the gear settling portion **142** of the moving device **140**.

(103) In the operation of driving the moving device (operation **S340**), after engaging a parking brake of the main gear **240** and fixing the main gear **240** to prevent the rotation of the main gear **240**, the motor **145** of the moving device **140** may operate. The moving device **140** may move along the transfer rail **130** by an operation of the motor **145** and may transfer the aircraft **200**.

(104) In the operation of confirming a position of the aircraft (operation **S350**), it may be possible to confirm the position of the aircraft **200** and confirm whether a transfer of the aircraft **200** has been completed to a target location where the aircraft **200** can be manually moved to the transfer area **113**. The target position can refer to a position of the aircraft **200** in a state in which the moving device **140** is moved to an end of the transfer rail **130** connected to the transfer area **113**. As a result of confirming the position of the aircraft **200**, when an additional transfer is necessary/desired, the moving device **140** can continue to be driven, and when the transfer to the target position is completed, the driving of the moving device **140** may be stopped.

(105) In the operation of moving the aircraft to the floor (operation **S360**), when the transfer of the aircraft **200** to the target location is completed and the driving of the moving device **140** stops, the parking brake of the main gear **240** can be released, and the ground personnel manually can move the aircraft **200** from the moving device **140** to the transfer area **113**.

(106) In the operation of driving a turntable (operation **S370**), a direction of the aircraft **200** may be adjusted by rotating the transfer area **113** in which the aircraft **200** is located.

(107) Although the example embodiment of the present disclosure has been described in detail above, the scope of the present disclosure is not necessarily limited thereto, and various modifications and variation may be made without departing from the technical concepts of the present disclosure described in the claims.

(108) Furthermore, in embodiments of the present disclosure, some components may be implemented in a deleted state, and the configurations of each embodiment may be configured in combination with each other.

Claims

1. A vertiport system comprising: a first area; a second area in a different location from the first area; and a transfer unit configured to transfer an aircraft between the first area and the second area, wherein the transfer unit comprises: at least one transfer rail connecting the first area and the

second area, and at least one moving device configured to accommodate at least a portion of a landing gear of the aircraft and configured to move the aircraft along the at least one transfer rail, wherein the at least one moving device comprises: a first moving device configured to accommodate a first main gear of the landing gear, and a second moving device configured to accommodate a second main gear of the landing gear, and wherein the at least one transfer rail comprises: a first rail configured to guide the first moving device, and a second rail configured to guide the second moving device.

2. The system of claim 1, wherein the at least one moving device comprises: a jig configured to connect to the landing gear, a wheel rotatably coupled to the jig, and a motor connected to the jig and configured to provide a driving force to rotate the wheel.

3. The system of claim 2, wherein the jig comprises: a gear settling portion configured to accommodate the landing gear, and a wheel connection portion to which the wheel is connected, and wherein the motor is positioned at the wheel connection portion and coupled to the wheel.

4. The system of claim 1, further comprising: a moving device controller configured to control an operation of the at least one moving device, wherein the at least one moving device further comprises a communication controller configured to receive a receiving signal from the moving device controller or transmit a transmitting signal to the moving device controller.

5. The system of claim 4, wherein the moving device controller is configured to generate a moving signal including at least one of or any combination of whether the at least one moving device is driven, a moving speed, and a moving direction, and wherein the moving device controller is further configured to transmit the moving signal to the communication controller.

6. The system of claim 5, wherein the communication controller is connected to a motor, and wherein the communication controller is configured to control driving of the motor based on the moving signal received from the moving device controller.

7. The system of claim 3, wherein the wheel connection portion is positioned below the gear settling portion, and wherein the gear settling portion comprises a settling groove having a tapered shape narrowing downwardly.

8. The system of claim 7, wherein the gear settling portion comprises: a bottom surface configured to accommodate the landing gear thereon, and a pair of guide surfaces extending obliquely outwardly in an upward direction from two sides of the bottom surface.

9. The system of claim 8, wherein the guide surfaces are configured to guide the landing gear to be aligned with the bottom surface during settling the landing gear in the gear settling portion due to a movement of the aircraft.

10. The system of claim 8, wherein the first area, the second area and the at least one transfer rail are provided on a floor of a port body, and wherein a rail accommodating groove recessed to a first depth is located in the floor such that the first rail or the second rail is positioned in the rail accommodating groove.

11. The system of claim 10, wherein the first depth of the rail accommodating groove and a respective first or second moving device is configured so that the bottom surface of the gear settling portion is positioned at a same height as the floor, in a state in which the respective first or second moving device is accommodated on the first rail or the second rail.

12. The system of claim 10, wherein the first depth of the rail accommodating groove and a respective first or second moving device is configured so that the bottom surface of the gear settling portion is positioned at a height lower than the floor by a specified length, in a state in which the respective first or second moving device is accommodated on the first rail or the second rail.

13. The system of claim 1, wherein the first rail and the second rail have a same shape and are spaced apart from each other at a first distance.

14. The system of claim 1, further comprising: a moving device controller, wherein each of the first and second rails has a layout including a straight section and a curved section, and wherein the

moving device controller is configured to control the first moving device and the second moving device so that the first moving device and the second moving device move at different speeds in the curved section.

15. The system of claim 13, wherein the system is configured so that, in a state in which the at least one moving device is positioned on the at least one transfer rail, the first and second main gears are accommodated in the first and second moving devices, respectively.

16. The system of claim 15, wherein a nose gear of the landing gear is in direct contact with a portion of a floor of a port body and positioned between the first rail and the second rail.

17. A vertiport system comprising: a first area; a second area in a different location from the first area; and a transfer unit configured to transfer an aircraft between the first area and the second area, wherein the transfer unit comprises: a transfer rail connecting the first area and the second area, and a moving device configured to accommodate at least a portion of a landing gear of the aircraft and configured to move the aircraft along the transfer rail, wherein the first area, the second area and the transfer rail are provided on a floor of a port body, wherein a rail accommodating groove recessed to a first depth is located in the floor such that the transfer rail is positioned in the rail accommodating groove, and wherein the first depth of the rail accommodating groove and the moving device are configured so that a bottom surface of a gear settling portion is positioned at a same height as the floor, in a state in which the moving device is accommodated on the transfer rail.

18. A vertiport system comprising: a first area; a second area in a different location from the first area; and a transfer unit configured to transfer an aircraft between the first area and the second area, wherein the transfer unit comprises: a transfer rail connecting the first area and the second area, and a moving device configured to accommodate at least a portion of a landing gear of the aircraft and configured to move the aircraft along the transfer rail, wherein the first area, the second area and the transfer rail are provided on a floor of a port body, wherein a rail accommodating groove recessed to a first depth is located in the floor such that the transfer rail is positioned in the rail accommodating groove, wherein the first depth of the rail accommodating groove and the moving device are configured so that a bottom surface of a gear settling portion is positioned at a height lower than the floor by a specified length, in a state in which moving device is accommodated on the transfer rail.
